

Can the model explain color without inventing damage?

Joint raises changed-vertex RMS on 30/36 mixtures. Passing thresholds: joint 3/6; geometry-only 2/6. No model is promoted.

This nonlearned experiment fits one dent and one surface-color patch separately. Healthy CAD, three possible centers, both event profiles, lighting and calibrated cameras are supplied.

Case group	Count	Role
Existing pure dents	18	Exposed development cases
Existing pure material patches	12	Six .8 dictionary-knot checks; six .6 cases
Existing healthy states	2	Zero physical change
New dent + material mixtures	36	12 same-center; 24 different-center

What ran

350 RGB renders · 204 frozen fields · 272 geometry records. Dictionary: 98 renders from healthy CAD. Mixtures: 252 actual Blender renders, created after dictionary freeze. Fit uses v004/v005 only; five other observed views are scored after all predictions freeze.

Fixed support threshold 10^{-5}

Model	Worst recall / 54 geometry cases	Worst false area / 12 material cases	Gate
Geometry only	87.690%	0.671%	FAIL
Appearance only	0.000%	0.000%	FAIL
Joint geometry + appearance	92.962%	0.000%	PASS
Unchanged CAD	0.000%	0.000%	FAIL

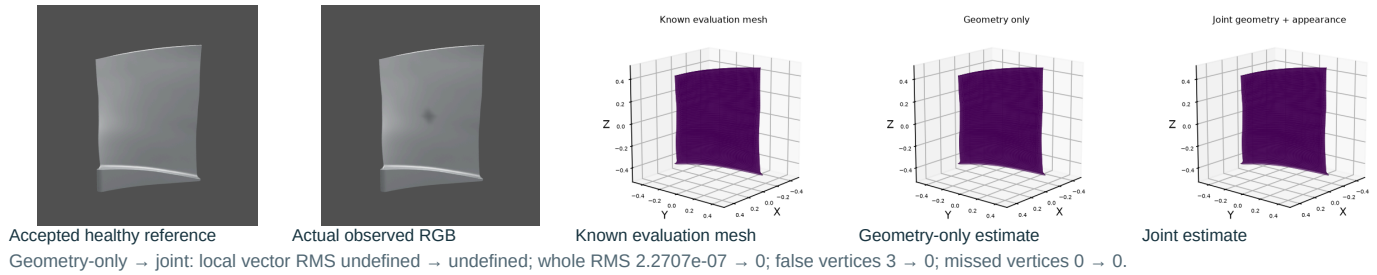
Every geometry-bearing case must retain $\geq 90\%$ eligible true-support area and every pure appearance case must have $\leq 1\%$ false physical-change area, at a common threshold. All six thresholds are retained. Coverage and physical shape error are separate from support recall.

Strong prior: at most one dent and one patch, each at one of three healthy-derived centers. This is a same-family, matched-renderer diagnostic on exposed sources 9/10, not arbitrary photograph-to-3D reconstruction or fresh-object validation.

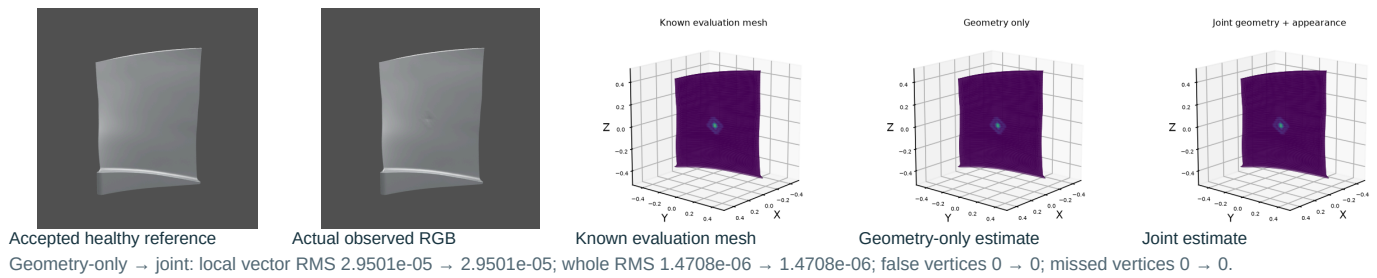
Pure appearance and pure physical damage

Fixed center-0 controls, shown for both objects. One common dent-side camera (azimuth40°, elevation15°) shows all meshes. Coordinates and colors share one scale; displacement is not magnified.

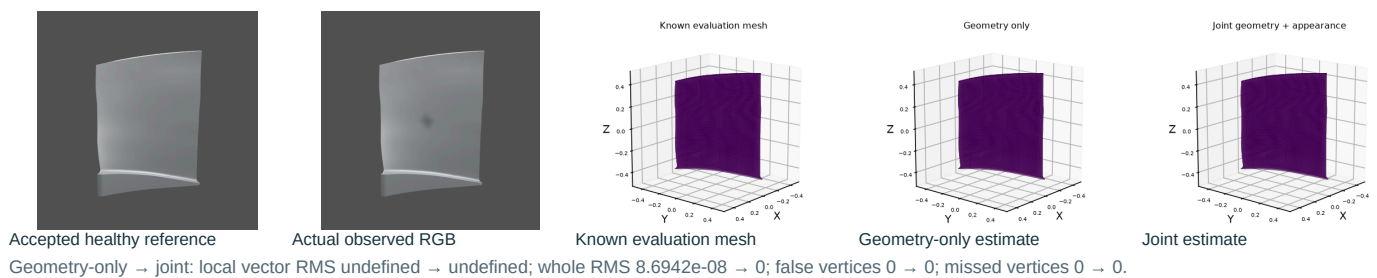
Source 9 · c021 · Pure material, center 0, multiplier .6



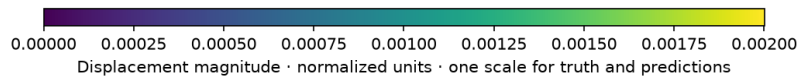
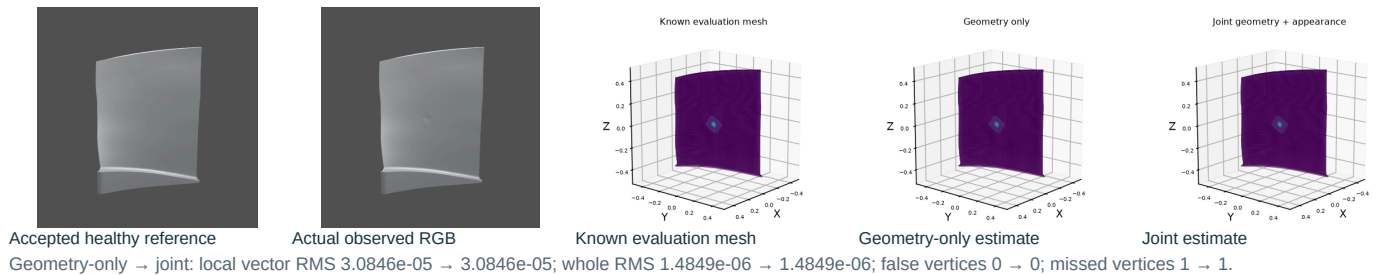
Source 9 · c003 · Pure dent, center 0, depth .002



Source 10 · c027 · Pure material, center 0, multiplier .6



Source 10 · c013 · Pure dent, center 0, depth .002

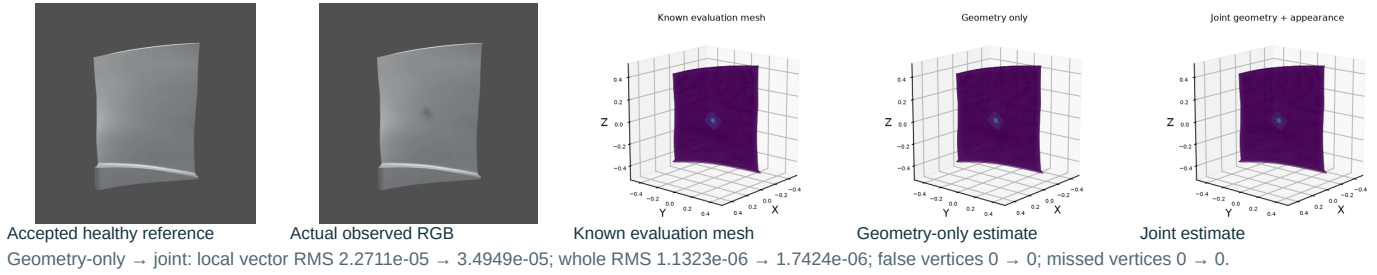


Truth is post-freeze evaluation only; color is displacement, not confidence. All airfoil faces are plotted; unchanged collar omitted. RGBs are byte-exact v005 observations. The archived healthy reference must match the fresh dictionary control within one channel level. None is an inferred-mesh re-render.

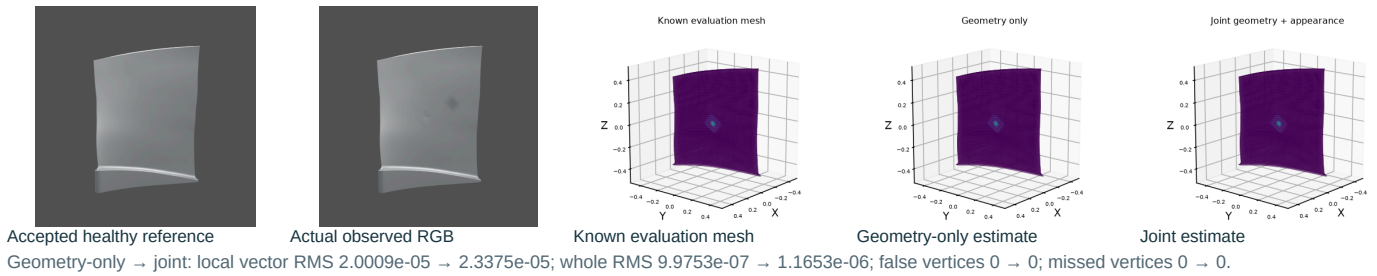
Two causes in the same observation

Fixed depth .0015 / multiplier .7 mixtures: same and different event centers. One common dent-side camera (azimuth40°, elevation15°) shows all meshes. Coordinates and colors share one scale; displacement is not magnified.

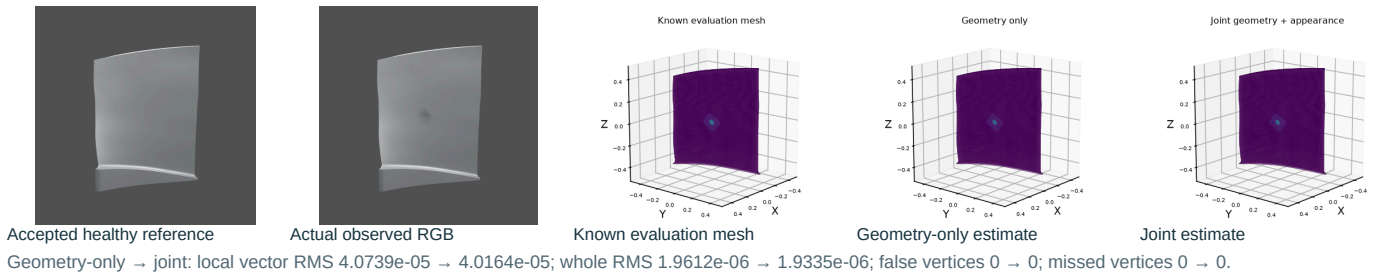
Source 9 · c033 · Same-center dent + material, depth .0015



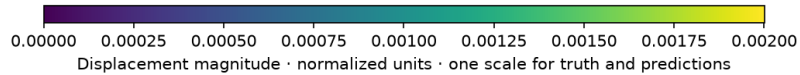
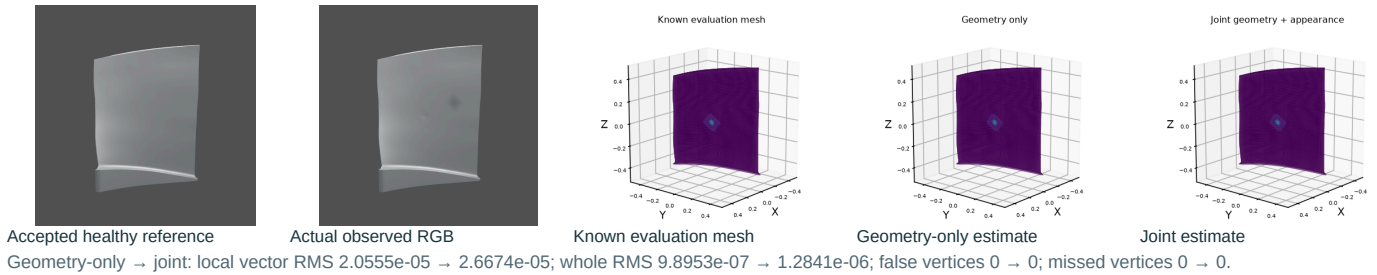
Source 9 · c035 · Different-center dent + material, depth .0015



Source 10 · c051 · Same-center dent + material, depth .0015

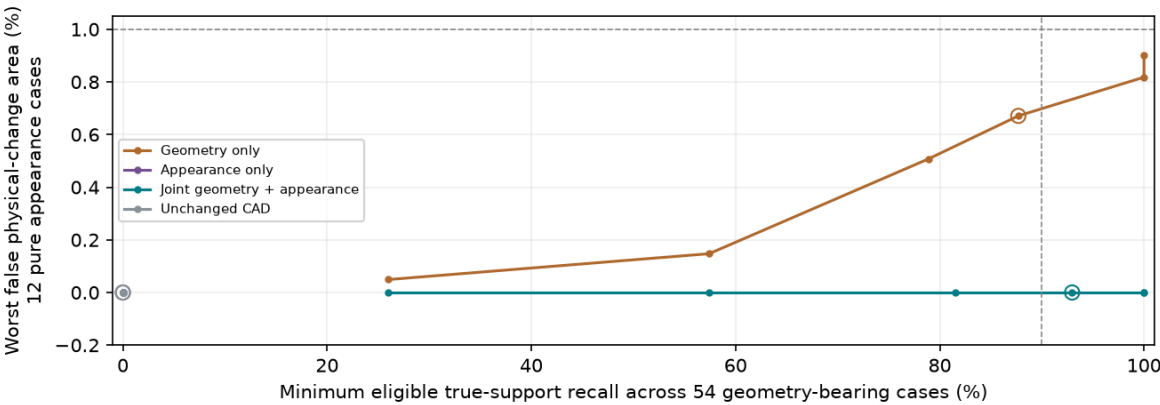


Source 10 · c053 · Different-center dent + material, depth .0015



Truth is post-freeze evaluation only; color is displacement, not confidence. All airfoil faces are plotted; unchanged collar omitted. RGBs are byte-exact v005 observations. The archived healthy reference must match the fresh dictionary control within one channel level. None is an inferred-mesh re-render.

Sensitivity and false physical change



Model	Threshold	Minimum recall / 54	Maximum false area / 12	Gate
Geometry only	2.5e-06	100.000%	0.901%	PASS
Geometry only	5e-06	100.000%	0.818%	PASS
Geometry only	1e-05	87.690%	0.671%	FAIL
Geometry only	2e-05	78.896%	0.508%	FAIL
Geometry only	4e-05	57.414%	0.147%	FAIL
Geometry only	8e-05	26.004%	0.049%	FAIL
Appearance only	2.5e-06	0.000%	0.000%	FAIL
Appearance only	5e-06	0.000%	0.000%	FAIL
Appearance only	1e-05	0.000%	0.000%	FAIL
Appearance only	2e-05	0.000%	0.000%	FAIL
Appearance only	4e-05	0.000%	0.000%	FAIL
Appearance only	8e-05	0.000%	0.000%	FAIL
Joint geometry + appearance	2.5e-06	100.000%	0.000%	PASS
Joint geometry + appearance	5e-06	100.000%	0.000%	PASS
Joint geometry + appearance	1e-05	92.962%	0.000%	PASS
Joint geometry + appearance	2e-05	81.481%	0.000%	FAIL
Joint geometry + appearance	4e-05	57.414%	0.000%	FAIL
Joint geometry + appearance	8e-05	26.004%	0.000%	FAIL
Unchanged CAD	2.5e-06	0.000%	0.000%	FAIL
Unchanged CAD	5e-06	0.000%	0.000%	FAIL
Unchanged CAD	1e-05	0.000%	0.000%	FAIL
Unchanged CAD	2e-05	0.000%	0.000%	FAIL
Unchanged CAD	4e-05	0.000%	0.000%	FAIL
Unchanged CAD	8e-05	0.000%	0.000%	FAIL

Open rings mark 10^{-5} . True support uses displacement norm $>10^{-5}$; predicted support uses norm strictly above each threshold. Exactly unchanged truth vertices define false movement. Empty eligible support is undefined and fails. Pure dent and mixture minima, each source, healthy results, whole-area false changes and coverage remain in the full supplements.

Does better image fit preserve the actual dent?

Joint regressions relative to geometry only

Group	N	Local RMS	Whole RMS	False	Missed	Faces
Pure dents	18	10/18	10/18	0/18	0/18	0/18
Pure material .8	6	undefined	0/6	0/6	0/6	0/6
Pure material .6	6	undefined	0/6	0/6	0/6	0/6
Healthy	2	undefined	0/2	0/2	0/2	0/2
Mixture same center	12	6/12	6/12	0/12	2/12	0/12
Mixture different centers	24	24/24	24/24	0/24	1/24	0/24
Mixture depth .00075	18	15/18	15/18	0/18	1/18	0/18
Mixture depth .0015	18	15/18	15/18	0/18	2/18	0/18

Counts are worse/defined cases using exact saved values. Groups overlap where explicitly labeled by depth. Appearance-only and unchanged CAD have zero physical displacement; they cannot preserve a real dent by construction.

Approximate RGB error, fitted versus unused views

Model	Fit MSE / 136 views	Unused MSE / 340 views	RGB values outside [0,1]
Geometry only	2.82e-06	5.9018e-07	0
Appearance only	2.3552e-07	1.7444e-08	0
Joint geometry + appearance	5.4748e-08	1.0247e-08	0
Unchanged CAD	3.0032e-06	5.9921e-07	0

Photo errors evaluate the un-clipped additive RGB approximation. These are not Blender rerenders of inferred geometry. The joint model is nested and has additional flexibility, so lower fitted-image error alone is not success.

Parameters and numerical ambiguity

Model	Depth MAE / 54	Material a MAE / 48	Dent center correct	Patch center correct
Geometry only	7.9516e-05	1.5	54/54	0/48
Appearance only	0.0011389	0.05293	0/54	48/48
Joint geometry + appearance	9.3905e-05	0.052303	54/54	48/48
Unchanged CAD	0.0011389	1.5	0/54	0/48

Model	Different tied geometry	Selected rank deficient	Max KKT
Geometry only	0/68	0/68	3.2266e-10
Appearance only	0/68	0/68	4.3999e-10
Joint geometry + appearance	0/68	0/68	4.4042e-10

Complete per-case centers, depth/material errors, all tied solutions, ranks, correlations, bound hits, conditioning and KKT residuals remain exported. Selected-tie KKT residuals are informational; the optimum and objective gap are checked separately. Ties are not calibrated uncertainty or proof of identifiability.

A controlled explanation, within a supplied family

Implemented and fixed

Healthy-only construction produces three geometry and three appearance columns. Geometry-only and appearance-only enumerate three centers; joint enumerates all nine center pairs. One-/two-variable box problems use fixed $[0,2]$ coefficient bounds, fixed numerical tolerances and declared tie rules. Geometry depth equals .0012g; material multiplier equals $1-.2a$. Only the geometry term moves the mesh.

What the observations can establish

Compare joint versus geometry-only using the same two observed views, dictionary and one-dent prior. Retain all 54 geometry-bearing cases, 12 pure material cases and two healthy cases. Reduced appearance-induced motion must be read alongside depth/vector error, retained support and shape diagnostics.

What remains unproven

- Stored PNGs use AgX. RGB addition is a finite-difference approximation, not linear radiance or an exact physical renderer.
- The dictionary and observations share a renderer and parametric event families. The three possible centers, profiles, radius, healthy CAD, lights and cameras are supplied.
- Sources 9/10 were previously exposed. Fresh mixtures are new conditions on those objects; reserved sources 11/12 remain unopened.
- Earlier learned models used seven views and different hypothesis classes. Their reports are context, not an equal-input performance comparison.
- Support localization does not establish accurate shape, metric real-world depth, generalization, inspection safety or novelty.
- Zero displacement is not abstention. Ranks/ties/KKT checks concern the numerical fit, not calibrated evidence confidence.
- Independent replay checks provenance/arithmetic/read chronology, not external replication. No model or threshold is promoted automatically.

Verification correction

The first verifier failed before outcome replay because it rejected matching $+\infty$ graph distances on disconnected collar vertices. A separately versioned checker fixes this comparison and performs the full replay. The failed receipt is retained; producer outputs, methods and numerical results were unchanged.

Complete portable records

Every saved result · 272 geometry metrics · 1,632 area rows · 24 threshold points · Per-view photo errors · Parameter errors · All fit diagnostics · Joint/geometry comparisons.

Each CSV has a full-precision JSON companion. See `SOURCE_MANIFEST.json` and `FIGURE_PROVENANCE.json`. PDF link actions are omitted; the HTML version uses relative links to companion files.

Independent receipt SHA256: 66a70f582e05df0c5764f9c20d48e9cced27a175edd804b75177c79ed4c76c5a